

CONTINUOUS INTERNAL EVALUATION - 2

Dept: AI/CD/CS	Sem / Div: 5 th	Sub: Theory of Computation	S Code: BCS503
Date: 19/12/2024	Time: 9:30-11:00	Max Marks: 50	Elective: N

Note: Answer any 2 full questions, choosing one full question from each part.

QN	Questions	Marks	RBT	CO's
PART A				
1 a	Define Deterministic PDA. Construct a PDA for $L = \{wCw^R \mid w \in \{a,b\}^*\}$ Draw graphical representation and write ID for babcbab and abcab	10	L2	CO3
b	Simplify the following grammar by eliminating useless symbols. $S \rightarrow aA \mid bB$ $A \rightarrow aA \mid a$ $B \rightarrow bB$ $D \rightarrow ab \mid Ea$ $E \rightarrow aC \mid d$	10	L3	CO3
c	Explain working of standard Turing machine along with formal definition and block diagram.	5	L2	CO4
OR				
2 a	i) Show that grammar $E \rightarrow E+E \mid E^*E \mid (E) \mid id$ is ambiguous for "id+id*id" ii) Convert the given CFG, $S \rightarrow aSb \mid ab$ to PDA.	10	L2	CO3

	b	Obtain LMD, RMD, Parse Tree for the string $+*-xyxy$ using the grammar $E \rightarrow +EE \mid *EE \mid -EE \mid x \mid y$	10	L3	CO3
	c	Explain Multitape Turing machine and non-deterministic TM.	5	L2	CO4
PART B					
3	a	Convert the following grammar into CNF. $S \rightarrow aACa$ $A \rightarrow B \mid a$ $B \rightarrow C \mid c$ $C \rightarrow Cc \mid \varepsilon$	10	L3	CO3
	b	Design a Turing machine for $L = \{a^n b^n : n \geq 1\}$. Write Transition diagram, transition table and ID for "aabb"	10	L3	CO4
	c	Prove that if M is a nondeterministic TM, there is a deterministic TM M1 such that $T(M) = T(M1)$	5	L2	CO5
OR					
4	a	Write the conditions of Pumping lemma. Prove that $L = \{a^n b^n c^n, n \geq 0\}$ is not context free using pumping lemma.	10	L3	CO3
	b	Design a Turing machine for $L = \{0^n 1^n 2^n : n \geq 1\}$. Write Transition diagram, transition table and ID for "001122"	10	L3	CO4
	c	Explain relationship between Recursive, Recursively Enumerable and non Recursively Enumerable languages with diagram.	5	L2	CO5

Prepared by: Roopa G K

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